

2024, Mar. 2024 Release LOC:Acute Adult

Infection: GI/GYN

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ^(1, 2)Episode Day 1 ⁽¹⁾Episode Day 2 ⁽¹¹⁰⁾Episode Day 3 ⁽¹⁴¹⁾

Episode Day 4-6

Episode Day 7

Discharge Screens ⁽¹⁴⁷⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for patients with suspected appendicitis, complicated appendicitis treated non-operatively, diverticulitis, pelvic inflammatory disease, *Clostridioides difficile* infection, spontaneous bacterial peritonitis, intra-abdominal abscess, pelvic abscess, tubo-ovarian abscess, acute (ascending) cholangitis, or toxic megacolon. For patients who begin to exhibit signs and symptoms consistent with sepsis and need additional interventions which are not provided in the current subset, please refer to the Infection: Sepsis subset.

Appendicitis, suspected or nonoperative management:

Clinical features of acute appendicitis are abdominal pain with either a fever, vomiting or nausea, or an elevated white blood cell count (WBC). When appendicitis is suspected clinically, but the imaging is indeterminate, follow-up re-examinations and repeat imaging may be necessary or appropriate for a definitive diagnosis (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187).

Complicated appendicitis with an abscess or phlegmon identified on imaging is routinely treated with laparoscopic appendectomy as first line approach, however, in selected patients, can be managed non-operatively with antibiotics and where available, percutaneous drainage (Di Saverio et al., World J Emerg Surg 2020, 15: 27). Since some abscesses may be challenging for percutaneous drainage or a complicated appendicitis treated nonoperatively could fail to respond to antibiotics and then recur, the role of surgery during the initial admission (emergency appendectomy, immediate appendectomy, or early appendectomy) becomes more important and may be indicated (Gavriilidis et al., J Clin Med Res 2019, 11: 56-64).

Acute (ascending) Cholangitis: Cholangitis is considered a gastrointestinal emergency requiring immediate diagnosis and treatment. It is characterized by acute inflammation of the biliary tree, usually caused by a gram-negative bacterial infection from biliary obstruction (commonly a bile duct stone), stasis, or mass effect, possibly from an adjacent tumor. The classic signs and symptoms of cholangitis are abdominal pain, jaundice, and fever (Charcot's Triad), although most patients do not present with all three simultaneously. Other symptoms may include mental status change or septic shock. Right upper quadrant ultrasound is usually the first-line imaging test. CT abdomen could also be performed. A magnetic resonance cholangiopancreatography (MRCP) can be ordered to define the biliary tree anatomy better. Treatment includes emergent parenteral broad-spectrum antibiotics, IV fluids resuscitation, and an endoscopic retrograde cholangiopancreatography (ERCP) for diagnosis and therapeutics in relieving and decompressing the bile duct (An et al., Gastroenterol Clin North Am 2021, 50: 403-14; Buxbaum et al., Gastrointest Endosc 2021, 94: 207-21.e14).

***Clostridioides difficile* infection (CDI):** CDI occurs when the colon is infected with the *C. difficile* toxin causing watery

diarrhea and an acute inflammatory process. Diagnosis is made by a positive stool test (PCR) for *C. difficile* or when pseudomembranes are found during colonoscopy. Treatment is dependent on the episode and clinical presentation. An initial episode is treated with oral fidaxomicin or vancomycin. Recurrent episodes may be treated with standard or pulsed vancomycin or oral fidaxomicin. When a recurrent episode presents within 6 months of a CDI, bezlotoxumab is recommended as an adjunct treatment due to favorable outcomes. Fulminant CDI is treated with oral vancomycin, however in cases with presence of an ileus, adding parenteral metronidazole to either oral or rectal vancomycin is recommended. If available, fecal microbiota transplantation (FMT), the process by which fecal material from a healthy donor is transferred to a recipient via a colonoscopic procedure may be considered in select patients who suffered two or more recurrences of CDI. Use of opiates or antidiarrheal medications should be avoided because of decreased intestinal motility which can worsen CDI and could precipitate toxic megacolon. Toxic megacolon may require urgent colectomy (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44; Kelly et al., Am J Gastroenterol 2021, 116: 1124-47; Poylin et al., Dis Colon Rectum 2021, 64: 650-68; McDonald et al., Clin Infect Dis 2018).

Diverticulitis: Acute diverticulitis is an infection of small bulging sacs (diverticula) of the inner lining of the bowel; it can be uncomplicated or complicated. Computed tomography (CT) is the preferred imaging for the diagnosis of diverticulitis, as it is highly sensitive and specific. Uncomplicated diverticulitis is confirmed when the CT shows localized inflammation or thickening of the colon wall whereas an abscess or perforation implies complicated disease. Most patients with uncomplicated diverticulitis can be managed as an outpatient with bowel rest and diet modifications (e.g., fluids, clear liquid diet). Treatment with an antibiotic is still recommended for patients who have significant comorbidities, signs of systemic inflammatory response, or those at risk for progressing to complicated diverticulitis (Qaseem et al., Ann Intern Med 2022, 175: 399-415; Sartelli et al., World J Emerg Surg 2020, 15: 32; Hall et al., Dis Colon Rectum 2020, 63: 728-47). When a patient has failed outpatient therapy, a CT can further diagnose complications, such as, an abscess or a perforation (Peery et al., Gastroenterology 2021, 160: 906-11.e1).

Abscess (intra-abdominal or pelvic):

Patients with an intra-abdominal or pelvic abscess typically present with a sudden onset of abdominal pain along with gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea). A fever may or may not be present. Computed tomography (CT) is the preferred diagnostic imaging technique due to high accuracy. The size and location of the abscess will determine whether treatment with intravenous anti-infectives alone is sufficient (e.g., for smaller fluid collections) and/or when drainage may be appropriate. If symptoms worsen or do not improve with antibiotics alone, drainage may subsequently be needed (Thompson and Hrabe, Gastroenterol Clin North Am 2021, 50: 475-88; Workowski et al., MMWR Recomm Rep 2021, 70: 1-187).

Pelvic Inflammatory Disease (PID):

Pelvic inflammatory disease (PID) is an inflammatory disorder of the upper genital tract usually caused by an infectious organism, and can be comprised of endometritis, salpingitis, tubo-ovarian abscess, and/or pelvic peritonitis.

Presenting symptoms may include abdominal or pelvic pain, fever, chills, abnormal vaginal discharge, or vomiting. Hospitalization is indicated when a surgical emergency (e.g., appendicitis) cannot be excluded, in patients who are pregnant, those who have features of severe disease, or in those who have failed outpatient management. Treatment with an antibiotic is essential to prevent long-term sequelae such as tubal occlusion, ectopic pregnancy, and infertility (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187; Curry et al., Am Fam Physician 2019, 100: 357-64).

Tubo-ovarian abscess (TOA) is an inflammatory mass involving the fallopian tube, ovary, and surrounding structures. TOA often occurs as a severe complication of pelvic inflammatory disease and presenting symptoms are similar. The preferred treatment of TOA is intravenous broad-spectrum anti-infectives and in most cases drainage. If medical management alone is not sufficient and symptoms persist or worsen (within 72 hours), then abscess drainage (ultrasound-guided or computed tomography-guided drainage) is required (if surgical drainage is required use General Surgical subset) (DeCesare et al., Am Surg 2021, 87: 120-4; Goje et al., J Minim Invasive Gynecol 2021, 28: 556-64; Weiss et al., J Am Coll Radiol 2020, 17: S265-s80).

Spontaneous Bacterial Peritonitis (SBP):

SBP is the abrupt onset of bacterial peritonitis in the absence of a recognizable cause of secondary peritonitis (e.g., perforation). It is diagnosed when ascitic fluid polymorphonuclear leukocyte (PMN) count is greater than 250 c.mm(25x10⁹/L). Patients may present with signs of fever, abdominal pain, nausea, encephalopathy, and deterioration in renal function. Treatment for SBP includes intravenous antimicrobials and volume expansion with albumin (Biggins et al., Hepatology 2021, 74: 1014-48; Mattos et al., Ann Hepatol 2020, 19: 451-7; Devani et al., J Clin Gastroenterol 2019, 53: e68-e74).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data

sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources:

- Appendicitis (Moris et al., *Jama* 2021, 326: 2299-311; Di Saverio et al., *World J Emerg Surg* 2020, 15: 27; Gavriilidis et al., *J Clin Med Res* 2019, 11: 56-64)
- Acute (ascending) Cholangitis (An et al., *Gastroenterol Clin North Am* 2021, 50: 403-14; Buxbaum et al., *Gastrointest Endosc* 2021, 94: 207-21.e14; Miura et al., *J Hepatobiliary Pancreat Sci* 2018, 25: 31-40)
- *Clostridium difficile* (Kelly et al., *Am J Gastroenterol* 2021, 116: 1124-47; Poylin et al., *Dis Colon Rectum* 2021, 64: 650-68; Johnson et al., *Clin Infect Dis* 2021, 73: e1029-e44; McDonald et al., *Clin Infect Dis* 2018)
- Diverticulitis (Qaseem et al., *Ann Intern Med* 2022, 175: 399-415; Peery et al., *Gastroenterology* 2021, 160: 906-11.e1; Hall et al., *Dis Colon Rectum* 2020, 63: 728-47; Sartelli et al., *World J Emerg Surg* 2020, 15: 32)
- Intra-abdominal or pelvic abscess (Workowski et al., *MMWR Recomm Rep* 2021, 70: 1-187; Thompson and Hrabe, *Gastroenterol Clin North Am* 2021, 50: 475-88)
- Pelvic Inflammatory Disease (includes tubo-ovarian abscess) (Workowski et al., *MMWR Recomm Rep* 2021, 70: 1-187)
- Spontaneous Bacterial Peritonitis (Biggins et al., *Hepatology* 2021, 74: 1014-48; Mattos et al., *Ann Hepatol* 2020, 19: 451-7; Devani et al., *J Clin Gastroenterol* 2019, 53: e68-e74)

(Excludes PO medications unless noted)

Select Day, One:

● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽³⁾

● Appendicitis, suspected and, **All:** ⁽⁴⁾

– Abdominal pain ⁽⁵⁾

● Finding, ≥ **One:**

– Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾

– Vomiting or nausea ⁽⁷⁾

– WBC ≥ 10,000/cu.mm($10 \times 10^9/L$) ⁽⁸⁾

– Imaging non-diagnostic ^(9, 10, 11)

● *C. difficile* infection and, **Both:** ^(12, 13)

● Finding, ≥ **One:**

– Continued deterioration despite ≥ 24h anti-infective (includes PO) treatment ⁽¹⁴⁾

– Noncompliant with outpatient anti-infective (includes PO)

● Unresponsive to outpatient anti-infective (includes PO) and, ≥ **One:** ^(15, 16)

– ≥ 3d

– ≥ 5 doses

● Dehydration, ≥ **One:**

● At least 1L IV fluid given prior to decision to admit and, ≥ **One:**

– Heart rate, sustained > 100/min ⁽¹⁷⁾

– Mental status changes (excludes coma, stupor, or obtundation) ⁽¹⁸⁾

● Orthostatic hypotension, **Both:** ^(19, 20)

– Sustained drop in blood pressure within 3 min of sitting or standing

– Systolic BP drop ≥ 20 mmHg or diastolic BP drop ≥ 10 mmHg

– Sodium > 150 mEq/L(150 mmol/L) ⁽²¹⁾

● Intervention, **Both:**

– Anti-infective (includes PO)

– IV fluid ⁽²²⁾

● Diverticulitis (uncomplicated) confirmed by CT and, **Both:** ^(23, 24, 25)

● Finding, ≥ **One:**

● High risk for disease progression and unable to tolerate oral intake and, ≥ **One:** ⁽²⁶⁾

– Severe systemic disease ^(27, 28)

– CRP > 14mg/dL(140 mg/L)

– Pain score > 7 ⁽²⁹⁾

– Symptoms, persistent > 5d ^(30, 31)

– Vomiting, persistent ⁽⁷⁾

– WBC ≥ 15,000/cu.mm($15 \times 10^9/L$)

● SIRS, ≥ **Two:** ⁽³²⁾

– Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽⁶⁾

● WBC, ≥ **One:**

– > 12,000/cu.mm($12 \times 10^9/L$) or < 4,000/cu.mm($4 \times 10^9/L$)

– Bands > 10%(0.10)

– Heart rate > 90/min, sustained ⁽¹⁷⁾

– Respiratory rate > 20/min, sustained ⁽¹⁷⁾

● Intervention, **Both:**

– Anti-infective ⁽²⁶⁾

– NPO and IV fluid ⁽²²⁾

● Pelvic inflammatory disease and, ≥ **One:** ⁽³³⁾

– Continued deterioration despite ≥ 24h treatment (includes PO) ⁽³⁴⁾

– Unable to tolerate oral anti-infective ⁽³⁴⁾

– Surgical emergency cannot be excluded ⁽³⁵⁾

● **ACUTE, ≥ One:** ⁽³⁶⁾

● Abscess, intra-abdominal or pelvic, **Both:** ⁽³⁷⁾

- Confirmed by imaging ^(38, 39)
- Anti-infective

● Appendicitis and nonoperative management, **Both:** ^(40, 41)

- Appendiceal abscess or phlegmon by imaging ^(9, 42)
- Anti-infective ⁽⁴³⁾

● Acute (ascending) cholangitis, actual or suspected and, **All:** ^(44, 45)

● Systemic inflammation, ≥ **One:** ^(44, 46)

- Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
- Shaking chills
- WBC ≥ 10,000/cu.mm(10x10⁹/L)
- WBC ≤ 4,000/cu.mm(4x10⁹/L)
- CRP ≥ 1mg/dL(10mg/L)
- Procalcitonin ≥ ULN ⁽⁴⁷⁾

● Finding, ≥ **One:**

● Cholestasis, ≥ **One:**

- Jaundice or bilirubin ≥ 2mg/dL(34.2μmol/L)
- Alkaline phosphatase (ALP) > 1.5x ULN
- Alanine transaminase (ALT) > 1.5x ULN
- Aspartate aminotransferase (AST) > 1.5x ULN

● Imaging finding, ≥ **One:** ^(48, 49)

- Biliary dilation
- Biliary stone, stricture, or stent
- Obstructing mass

– Anti-infective ⁽⁵⁰⁾

● *C. difficile* infection and, **Both:** ^(13, 51)

● Finding, ≥ **One:** ⁽⁵¹⁾

- Temperature > 101.3°F(38.5°C) ⁽⁵²⁾
- WBC ≥ 15,000/cu.mm(15x10⁹/L)
- Creatinine ≥ 1.5x baseline or estimated baseline ^(53, 54)

● Intervention, **Both:**

- Anti-infective (includes PO)
- IV fluid ⁽²²⁾

● Diverticulitis and, **Both:** ⁽²⁵⁾

● Finding, ≥ **One:**

- Complicated CT finding ^(55, 56, 57, 58)

● Comorbid, ≥ **One:** ⁽⁵⁹⁾

- End stage liver disease
- Immunocompromised ⁽⁶⁰⁾
- Renal disease and chronic dialysis

● Organ dysfunction or hypoperfusion, ≥ **One:** ⁽⁶¹⁾

● Moderate risk finding, ≥ **Two:**

- Creatinine ≥ 1.5x and < 2x baseline or estimated baseline ⁽⁵³⁾
- Mental status changes (excludes coma, stupor, or obtundation) and GCS 13-14 ⁽¹⁸⁾
- Platelet count 100,000/cu.mm(100x10⁹/L) to 150,000/cu.mm(150x10⁹/L) and < baseline ⁽⁵⁴⁾
- Total bilirubin 1.2 mg/dL(20.52 μmol/L) to 1.9 mg/dL(32.49 μmol/L) and > baseline ⁽⁵⁴⁾
- O₂ sat ≤ 91%(0.91) and < baseline ⁽⁵⁴⁾

● High risk finding, ≥ **One:**

- Creatinine ≥ 2x baseline or estimated baseline ⁽⁵³⁾
- Mental status changes (excludes coma, stupor, or obtundation) and GCS 10-12 ⁽¹⁸⁾
- Platelet count < 100,000/cu.mm(100x10⁹/L) and < baseline ⁽⁵⁴⁾

- Total bilirubin ≥ 2 mg/dL (34.2 μ mol/L) and $>$ baseline ⁽⁵⁴⁾
- Lactic acid $>$ ULN and < 4 mmol/L (36.0 mg/dL) ⁽⁶²⁾
- Intervention, $\geq$ **One**:
 - Anti-infective ⁽⁵⁷⁾
 - Abscess drainage ≤ 24 h ⁽⁶³⁾
- Pelvic inflammatory disease and, $\geq$ **One**: ^(33, 64)
 - Pregnancy ⁽⁶⁵⁾
- Severe illness, $\geq$ **One**:
 - Temperature $> 101^\circ\text{F}$ (38.3°C) ⁽⁶⁶⁾
 - Vomiting, protracted
 - Tubo-ovarian abscess confirmed by imaging ^(67, 68)
- Persistent symptoms despite outpatient anti-infective (includes PO) treatment ≥ 3 d and, $\geq$ **One**: ⁽⁶⁹⁾
 - Abdominal pain, persistent
 - Temperature $\geq 100.4^\circ\text{F}$ (38.0°C) ⁽⁶⁾
- SBP and, **All**: ⁽⁷⁰⁾
 - Polymorphonuclear leukocyte count $> 250/\text{cu.mm}$ (0.25x10⁹/L)
 - Albumin ⁽⁷¹⁾
 - Anti-infective ⁽⁷²⁾
- INTERMEDIATE, $\geq$ **One**: ⁽⁷³⁾
 - SBP and hepatic encephalopathy Grade III and, $\geq$ **One**: ^(70, 74)
 - Confusion
 - Gross disorientation ⁽⁷⁵⁾
 - Somnolence to semi-stupor and responsive to stimuli ⁽⁷⁶⁾
 - Unusual or bizarre behavior ⁽⁷⁷⁾
 - HFNC and respiratory monitoring every 3-4h ^(78, 79)
 - NIPPV ⁽⁸⁰⁾
 - Oxygen $\geq 40\%$ (0.40), ≤ 2 d ⁽⁸¹⁾
- CRITICAL, $\geq$ **One**: ⁽⁸²⁾
 - Fulminant *C. difficile* infection and, **Both**: ^(83, 84)
 - Ileus ⁽⁸⁵⁾
 - IV fluid resuscitation ≤ 2 d ⁽⁸⁶⁾
 - Severe cholangitis and, **Both**: ⁽⁸⁷⁾
 - Organ dysfunction, $\geq$ **Two**: ⁽⁸⁸⁾
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 10-12 ⁽⁸⁹⁾
 - INR > 1.5 and $>$ baseline without anticoagulation ⁽⁹⁰⁾
 - PaO₂/FiO₂ < 300 mmHg (39.9 kPa)
 - Platelet count $< 100,000/\text{cu.mm}$ (100x10⁹/L) ⁽⁹¹⁾
 - Serum creatinine ≥ 2 x baseline or estimated baseline ⁽⁵³⁾
 - Biliary decompression or drainage ≤ 24 h ⁽⁹²⁾
 - SBP and hepatic encephalopathy Grade IV ^(74, 93)
 - Toxic megacolon confirmed by imaging and, $\geq$ **One**: ^(94, 95, 96)
 - Blood product transfusion
 - Colonic decompression
 - IV fluid resuscitation ≤ 2 d ⁽⁸⁶⁾
 - CRRT ⁽⁹⁷⁾
 - DIC ⁽⁹⁸⁾
 - ECMO or ECLS ^(99, 100)
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both**: ⁽¹⁰¹⁾
 - Hypotension, $\geq$ **One**:
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One**:

- Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽¹⁰²⁾
- Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽¹⁰³⁾
- HFNC and respiratory monitoring every 1-2h ^(78, 79, 81)
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽⁸⁰⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One**:
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure) ^(104, 105)
- **Episode Day 1, One**: ⁽¹⁾
(Symptom or finding within 24h)
- **OBSERVATION, ≥ One**: ⁽³⁾
 - Appendicitis, suspected and, **All**: ⁽⁴⁾
 - Abdominal pain ⁽⁵⁾
 - Finding, ≥ **One**:
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - Vomiting or nausea ⁽⁷⁾
 - WBC ≥ 10,000/cu.mm(10x10⁹/L) ⁽⁸⁾
 - Imaging non-diagnostic ^(9, 10, 11)
 - Intervention, ≥ **One**:
 - Anti-infective
 - NPO and IV fluid ⁽²²⁾
- *C. difficile* infection and, **Both**: ^(12, 13)
 - Finding, ≥ **One**:
 - Continued deterioration despite ≥ 24h anti-infective (includes PO) treatment ⁽¹⁴⁾
 - Noncompliant with outpatient anti-infective (includes PO)
 - Unresponsive to outpatient anti-infective (includes PO) and, ≥ **One**: ^(15, 16)
 - ≥ 3d
 - ≥ 5 doses
 - Dehydration, ≥ **One**:
 - At least 1L IV fluid given prior to decision to admit and, ≥ **One**:
 - Heart rate, sustained > 100/min ⁽¹⁷⁾
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 13-14 ⁽¹⁸⁾
 - Orthostatic hypotension, **Both**: ^(19, 20)
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Systolic BP drop ≥ 20 mmHg or diastolic BP drop ≥ 10 mmHg
 - Sodium > 150 mEq/L(150 mmol/L) ⁽²¹⁾
 - Intervention, **Both**:
 - Anti-infective (includes PO)
 - IV fluid ⁽²²⁾
- Diverticulitis (uncomplicated) confirmed by CT and, **Both**: ^(23, 24, 25)
 - Finding, ≥ **One**:
 - High risk for disease progression and unable to tolerate oral intake and, ≥ **One**: ⁽²⁶⁾
 - Severe systemic disease ^(27, 28)
 - CRP > 14mg/dL(140mg/L)
 - Pain score > 7 ⁽²⁹⁾
 - Symptoms, persistent > 5 days ^(30, 31)
 - Vomiting, persistent ⁽⁷⁾
 - WBC ≥ 15,000/cu.mm(15x10⁹/L)
 - SIRS, ≥ **Two**: ⁽³²⁾
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽⁶⁾

- **WBC, ≥ One:**
 - > 12,000/cu.mm($12 \times 10^9/L$) or < 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - Heart rate > 90/min, sustained ⁽¹⁷⁾
 - Respiratory rate > 20/min, sustained ⁽¹⁷⁾
- **Intervention, Both:**
 - Anti-infective ⁽²⁶⁾
 - NPO and IV fluid ⁽²²⁾
- **Pelvic inflammatory disease and, Both:** ⁽³³⁾
 - **Finding, ≥ One:**
 - Continued deterioration despite ≥ 24h treatment (includes PO) ⁽³⁴⁾
 - Unable to tolerate oral anti-infective ⁽³⁴⁾
 - Surgical emergency cannot be excluded ⁽³⁵⁾
 - Anti-infective ⁽¹⁰⁶⁾
- **ACUTE, ≥ One:** ⁽³⁶⁾
 - Appendicitis and pregnant (**use General Surgical subset**) ⁽¹⁰⁷⁾
 - **Abscess, intra-abdominal or pelvic, Both:** ⁽³⁷⁾
 - Confirmed by imaging ^(38, 39)
 - Anti-infective
 - **Appendicitis and nonoperative management, Both:** ⁽⁴⁰⁾
 - Appendiceal abscess or phlegmon by imaging ^(9, 42)
 - Anti-infective ⁽⁴³⁾
 - **Acute (ascending) cholangitis, actual or suspected and, All:** ^(44, 45, 46)
 - **Systemic inflammation, ≥ One:** ⁽⁴⁶⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - Shaking chills
 - WBC ≥ 10,000/cu.mm($10 \times 10^9/L$)
 - WBC ≤ 4,000/cu.mm($4 \times 10^9/L$)
 - CRP ≥ 1mg/dL(10mg/L)
 - Procalcitonin ≥ ULN ⁽⁴⁷⁾
 - **Finding, ≥ One:**
 - **Cholestasis, ≥ One:**
 - Jaundice or bilirubin ≥ 2mg/dL($34.2 \mu\text{mol/L}$)
 - Alkaline phosphatase (ALP) > 1.5x ULN
 - Alanine transaminase (ALT) > 1.5x ULN
 - Aspartate aminotransferase (AST) > 1.5x ULN
 - **Imaging finding, ≥ One:** ^(48, 49)
 - Biliary dilation
 - Biliary stone, stricture, or stent
 - Obstructing mass
 - Anti-infective ⁽⁵⁰⁾
 - **C. difficile infection and, Both:** ^(13, 51)
 - **Finding, ≥ One:** ⁽⁵¹⁾
 - Temperature > 101.3°F(38.5°C) ⁽⁵²⁾
 - WBC ≥ 15,000/cu.mm($15 \times 10^9/L$)
 - Creatinine ≥ 1.5x baseline or estimated baseline ^(53, 54)
 - Hospital acquired during inpatient hospitalization ⁽¹⁰⁸⁾
 - **Intervention, Both:**
 - Anti-infective (includes PO)
 - IV fluid ⁽²²⁾
 - **Diverticulitis and, Both:** ⁽²⁵⁾
 - **Finding, ≥ One:**
 - Complicated CT finding ^(55, 56, 57, 58)

- Comorbid, ≥ **One:** ^(59, 109)
 - End stage liver disease
 - Immunocompromised ⁽⁶⁰⁾
 - Renal disease and chronic dialysis
- Organ dysfunction or hypoperfusion, ≥ **One:** ⁽⁶¹⁾
 - Moderate risk finding, ≥ **Two:**
 - Creatinine ≥ 1.5x and < 2x baseline or estimated baseline ⁽⁵³⁾
 - Mental status changes (excludes coma, stupor, or obtundation) and GCS 13-14 ⁽¹⁸⁾
 - Platelet count 100,000/cu.mm(100x10⁹/L) to 150,000/cu.mm(150x10⁹/L) and < baseline ⁽⁵⁴⁾
 - Total bilirubin 1.2 mg/dL(20.52 μmol/L) to 1.9 mg/dL(32.49 μmol/L) and > baseline ⁽⁵⁴⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽⁵⁴⁾
 - High risk finding, ≥ **One:**
 - Creatinine ≥ 2x baseline or estimated baseline ⁽⁵³⁾
 - Mental status changes (excludes coma, stupor, or obtundation) and GCS 10-12 ⁽¹⁸⁾
 - Platelet count < 100,000/cu.mm(100x10⁹/L) and < baseline ⁽⁵⁴⁾
 - Total bilirubin ≥ 2 mg/dL(34.2 μmol/L) and > baseline ⁽⁵⁴⁾
 - Lactic acid > ULN and < 4 mmol/L(36.0 mg/dL) ⁽⁶²⁾
- Intervention, ≥ **One:**
 - Anti-infective ⁽⁵⁷⁾
 - Abscess drainage ≤ 24h ⁽⁶³⁾
- Pelvic inflammatory disease and, **Both:** ^(33, 64)
 - Finding, **One:**
 - Pregnancy ⁽⁶⁵⁾
 - Severe illness, ≥ **One:**
 - Temperature > 101°F(38.3°C) ⁽⁶⁶⁾
 - Vomiting, protracted
 - Tubo-ovarian abscess confirmed by imaging ^(67, 68)
 - Persistent symptoms despite outpatient anti-infective (includes PO) treatment ≥ 3d and, ≥ **One:** ⁽⁶⁹⁾
 - Abdominal pain, persistent
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - Anti-infective ⁽¹⁰⁶⁾
- SBP and, **All:** ⁽⁷⁰⁾
 - Polymorphonuclear leukocyte count > 250/cu.mm(0.25x10⁹/L)
 - Albumin ⁽⁷¹⁾
 - Anti-infective ⁽⁷²⁾
- **INTERMEDIATE**, ≥ **One:** ⁽⁷³⁾
 - SBP and hepatic encephalopathy Grade III and, ≥ **One:** ^(70, 74)
 - Confusion
 - Gross disorientation ⁽⁷⁵⁾
 - Somnolence to semi-stupor and responsive to stimuli ⁽⁷⁶⁾
 - Unusual or bizarre behavior ⁽⁷⁷⁾
 - HFNC and respiratory monitoring every 3-4h ^(78, 79)
 - NIPPV
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁸¹⁾
- **CRITICAL**, ≥ **One:** ⁽⁸²⁾
 - Fulminant *C. difficile* infection and, **Both:** ^(83, 84)
 - Ileus ⁽⁸⁵⁾
 - IV fluid resuscitation ≤ 2d ⁽⁸⁶⁾
 - Severe cholangitis and, **Both:** ⁽⁸⁷⁾
 - Organ dysfunction, ≥ **Two:** ⁽⁸⁸⁾
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 10-12 ⁽⁸⁹⁾
 - INR > 1.5 and > baseline without anticoagulation ⁽⁹⁰⁾
 - PaO₂/FiO₂ < 300 mmHg(39.9 kPa)

- Platelet count < 100,000/cu.mm(100x10⁹/L) ⁽⁹¹⁾
- Serum creatinine ≥ 2x baseline or estimated baseline ⁽⁵³⁾
- Biliary decompression or drainage ≤ 24h ⁽⁹²⁾
- SBP and hepatic encephalopathy Grade IV ^(74, 93)
- Toxic megacolon confirmed by imaging and, ≥ **One:** ^(94, 95, 96)
 - Blood product transfusion
 - Colonic decompression
 - IV fluid resuscitation ≤ 2d ⁽⁸⁶⁾
- CRRT ⁽⁹⁷⁾
- DIC ⁽⁹⁸⁾
- ECMO or ECLS ^(99, 100)
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** ⁽¹⁰¹⁾
 - Hypotension, ≥ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Finding, ≥ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽¹⁰²⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽¹⁰³⁾
- HFNC and respiratory monitoring every 1-2h ^(78, 79, 81)
- Mechanical ventilation
- NIPPV and, ≥ **One:** ⁽⁸⁰⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure) ^(104, 105)
- **Episode Day 2, One:** ⁽¹¹⁰⁾
 - **OBSERVATION, One:**
 - Appendicitis, suspected and, **One:** ⁽¹¹¹⁾
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾
 - Afebrile
 - Appendicitis ruled out
 - Laboratory values within acceptable limits ⁽¹¹³⁾
 - Pain controlled or manageable ⁽¹¹⁴⁾
 - Tolerating PO ⁽¹¹⁵⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For operative management of appendicitis (**use General Surgical subset**)
 - For a condition other than appendicitis use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **C. difficile infection and, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾
 - Anti-infective regimen established and tolerated
 - Decrease in frequency and severity of diarrhea
 - Laboratory values within acceptable limits ⁽¹¹³⁾
 - Mental status or GCS within normal limits or returned to baseline
 - Tolerating PO ⁽¹¹⁵⁾
 - Vital signs within acceptable limits ⁽¹¹³⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - **C. difficile infection and, Both:**

- Worsening frequency and severity of diarrhea
- **Intervention, Both:**
 - Anti-infective (includes PO)
 - Continuous IV fluid replacement based on losses ⁽¹¹⁶⁾
- **Diverticulitis and, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾
 - Afebrile
 - Laboratory values within acceptable limits ⁽¹¹³⁾
 - Vital signs within acceptable limits ⁽¹¹³⁾
 - Pain controlled or manageable ⁽¹¹⁴⁾
 - Tolerating PO ⁽¹¹⁵⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - **Diverticulitis and, ≥ One:**
 - Anti-infective
 - Transition to PO anti-infective ≤ 24hr
- **Pelvic inflammatory disease and, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾
 - Afebrile
 - Anti-infective regimen established and tolerated
 - Pain controlled or manageable ⁽¹¹⁴⁾
 - Surgical emergency ruled out
 - Tolerating PO ⁽¹¹⁵⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - Pelvic inflammatory disease and anti-infective
- **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾
 - Afebrile
 - Anti-infective regimen established and tolerated
 - Laboratory values within acceptable limits ⁽¹¹³⁾
 - Tolerating PO ⁽¹¹⁵⁾
 - **Infection resolving and, ≥ One:** ⁽¹¹⁷⁾
 - Diverticulitis and pain controlled or manageable ⁽¹¹⁴⁾
 - **Pelvic inflammatory disease (includes tubo-ovarian abscess) and, ≥ One:**
 - Pain controlled or manageable ^(111, 114)
 - Drain removed or drainage manageable
 - No complication or active comorbidity relevant to this episode of care ⁽¹¹⁸⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Gastrointestinal (GI) Bleeding (**use Gastrointestinal (GI) Bleeding subset**)
 - Surgical intervention required (**use General Surgical subset**)
 - Abscess, intra-abdominal or pelvic and anti-infective ≤ 6d since initiation
 - Appendicitis (nonoperative management) and anti-infective ^(40, 119)
 - Acute (ascending) cholangitis and anti-infective ≤ 5d since initiation
 - Biliary decompression or drainage ≤ 24h
 - **C. difficile infection and, Both:**
 - Worsening frequency and severity of diarrhea
 - **Intervention, Both:**
 - Anti-infective (includes PO) ≤ 5d since initiation
 - Continuous IV fluid replacement based on losses ⁽¹¹⁶⁾
 - Diverticulitis and anti-infective ≤ 5d since initiation
 - **Pelvic inflammatory disease and anti-infective ≤ 4d since initiation and, ≥ One:**
 - Pregnancy ⁽¹²⁰⁾
 - Severe illness ⁽¹²¹⁾
 - Tubo-ovarian abscess ^(63, 122)

- Persistent symptoms, ≥ **One**:
 - Temperature ≥ 100.4°F(38.0°C)
 - Abdominal pain, persistent
- SBP and anti-infective ≤ 6d since initiation
- Abscess drainage requiring drain manipulation ≤ 24h ^(123, 124)
- Hepatic encephalopathy Grade II and, ≥ **One**: ^(74, 125, 126)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹²⁷⁾
 - Lactulose (includes PO or PR) ⁽¹²⁸⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽¹²⁹⁾
 - Neomycin (includes PO) ⁽¹³⁰⁾
 - Rifaximin (includes PO) ⁽¹³¹⁾
- Adverse anti-infective drug reaction requiring change in medication ≤ 24h ⁽¹³²⁾
- Blood glucose ≤ 24h, **All**: ⁽¹³³⁾
 - Finding, ≥ **One**:
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(133, 134)
- Ileus and, **One**: ⁽¹³⁵⁾
 - Surgical intervention required (**use General Surgical subset**)
 - Resolved and initiating PO or tube feeding ≤ 2d
 - Unresolved and IV fluid ≤ 5d ⁽²²⁾
- Inadequate oral intake and IV fluid ≤ 4d ^(22, 136)
- Post Critical care ≤ 24h
- **INTERMEDIATE**, ≥ **One**:
 - SBP and hepatic encephalopathy Grade III and, ≥ **One**: ^(70, 74)
 - Confusion
 - Gross disorientation ⁽⁷⁵⁾
 - Somnolence to semi-stupor and responsive to stimuli ⁽⁷⁶⁾
 - Unusual or bizarre behavior ⁽⁷⁷⁾
 - Hepatorenal syndrome and, ≥ **One**: ⁽¹³⁷⁾
 - Albumin
 - Octreotide and midodrine (includes PO)
 - HFNC and respiratory monitoring every 3-4h ^(78, 79)
 - NIPPV ⁽⁸⁰⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁸¹⁾
- **CRITICAL**, ≥ **One**:
 - Fulminant *C. difficile* infection and, **Both**: ⁽⁸³⁾
 - Ileus ⁽⁸⁵⁾
 - IV fluid resuscitation ≤ 2d ^(84, 86)
 - Severe cholangitis and biliary decompression or drainage ≤ 24h ^(87, 92)
 - SBP and hepatic encephalopathy Grade IV ^(74, 93)
 - Toxic megacolon and, ≥ **One**: ⁽⁹⁴⁾
 - Blood product transfusion
 - Colonic decompression
 - IV fluid resuscitation ≤ 2d ⁽⁸⁶⁾
 - Hepatorenal syndrome and, **Both**: ⁽¹³⁷⁾
 - Hemodynamic instability ⁽¹³⁸⁾
 - Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Continuous renal replacement therapy (CRRT) ^(97, 139)
 - CRRT ⁽⁹⁷⁾
 - DIC ⁽⁹⁸⁾

- ECMO or ECLS ⁽¹⁰⁰⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽¹⁰¹⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽¹⁰²⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽¹⁰³⁾
- HFNC and respiratory monitoring every 1-2h ^(78, 79, 81)
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽⁸⁰⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽¹⁴⁰⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 3, **One:** ⁽¹⁴¹⁾
 - OBSERVATION, **One:**
 - Responder, discharge expected today if clinically stable last 12h, $\geq$ **One:** ⁽¹¹²⁾
 - *C. difficile* infection and, **All:**
 - Anti-infective regimen established and tolerated
 - Decrease in frequency and severity of diarrhea
 - Tolerating PO ⁽¹¹⁵⁾
 - Diverticulitis and, **Both:**
 - Pain controlled or manageable ⁽¹¹⁴⁾
 - Tolerating PO ⁽¹¹⁵⁾
 - Pelvic inflammatory disease and, **All:**
 - Afebrile
 - Anti-infective regimen established and tolerated
 - Pain controlled or manageable ⁽¹¹⁴⁾
 - Tolerating PO ⁽¹¹⁵⁾
 - Non-responder, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - *C. difficile* infection, **One:**
 - For *C. difficile* infection use Episode Day 3 at a higher level of care
 - For a condition other than *C. difficile* infection use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - Diverticulitis, **One:**
 - For diverticulitis use Episode Day 3 at a higher level of care
 - For a condition other than diverticulitis use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - Pelvic inflammatory disease, **One:**
 - For pelvic inflammatory disease use Episode Day 3 at a higher level of care
 - For a condition other than pelvic inflammatory disease use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - ACUTE, **One:**
 - Responder, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾
 - Afebrile
 - Anti-infective regimen established and tolerated
 - Laboratory values within acceptable limits ⁽¹¹³⁾

- Tolerating PO ⁽¹¹⁵⁾
- Infection resolving and, ≥ **One:** ⁽¹¹⁷⁾
 - Acute (ascending) cholangitis and pain controlled or manageable ⁽¹¹⁴⁾
 - Appendicitis and, ≥ **One:** ^(111, 114)
 - Pain controlled or manageable ^(111, 114)
 - Drain removed or drainage manageable
 - *C. difficile* infection and decrease in frequency and severity of diarrhea
 - Diverticulitis and pain controlled or manageable ⁽¹¹⁴⁾
 - Pelvic inflammatory disease (includes tubo-ovarian abscess) and, ≥ **One:**
 - Pain controlled or manageable ^(111, 114)
 - Drain removed or drainage manageable
 - SBP and pain controlled or manageable ⁽¹¹⁴⁾
- Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹¹⁸⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
 - Adverse drug reaction resolving
 - Blood glucose within acceptable limits ⁽¹¹³⁾
 - Hepatic encephalopathy and, **Both:** ⁽⁷⁴⁾
 - Grade I or within acceptable limits ⁽¹¹³⁾
 - Neurological stability last 24h ⁽¹⁴²⁾
 - Ileus and, **One:**
 - Tolerating PO ⁽¹¹⁵⁾
 - Nutritional route established ⁽¹⁴³⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Gastrointestinal (GI) Bleeding (**use Gastrointestinal (GI) Bleeding subset**)
 - Surgical intervention required (**use General Surgical subset**)
 - Abscess, intra-abdominal or pelvic and anti-infective ≤ 6d since initiation
 - Appendicitis (nonoperative management) and anti-infective ^(40, 119)
 - Acute (ascending) cholangitis and anti-infective ≤ 5d since initiation
 - Biliary decompression or drainage ≤ 24h
 - *C. difficile* infection and, **Both:**
 - Worsening frequency and severity of diarrhea
 - Intervention, **Both:**
 - Anti-infective (includes PO) ≤ 5d since initiation
 - Continuous IV fluid replacement based on losses ⁽¹¹⁶⁾
 - Diverticulitis and anti-infective ≤ 5d since initiation and, ≥ **One:**
 - High risk comorbid ^(59, 144)
 - Temperature ≥ 100.4°F(38.0°C) ⁽⁶⁾
 - Unable to tolerate oral intake and, **Both:**
 - NPO
 - IV fluid ⁽²²⁾
 - WBC, ≥ **One:**
 - > 12,000/cu.mm(12x10⁹/L) or < 4,000/cu.mm(4x10⁹/L)
 - Bands > 10%(0.10)
 - ANC < 500/cu.mm(500x10⁶/L) ⁽¹⁴⁵⁾
 - Abscess drainage ≤ 24h ⁽⁶³⁾
 - Pelvic inflammatory disease and anti-infective ≤ 4d since initiation and, ≥ **One:**
 - Pregnancy ⁽¹²⁰⁾
 - Severe illness ⁽¹²¹⁾
 - Tubo-ovarian abscess ^(63, 122)
 - Persistent symptoms, ≥ **One:**
 - Temperature ≥ 100.4°F(38.0°C)
 - Abdominal pain, persistent

- SBP and anti-infective $\leq 6d$ since initiation
- Toxic megacolon and parenteral nutrition
- Abscess drainage requiring drain manipulation $\leq 24h$ ^(123, 124)
- Adverse anti-infective drug reaction requiring change in medication $\leq 24h$ ⁽¹³²⁾
- Blood glucose $\leq 24h$, **All:** ⁽¹³³⁾
 - Finding, $\geq$ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(133, 134)
- Hepatic encephalopathy Grade II and, $\geq$ **One:** ^(74, 125, 126)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹²⁷⁾
 - Lactulose (includes PO or PR) ⁽¹²⁸⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽¹²⁹⁾
 - Neomycin (includes PO) ⁽¹³⁰⁾
 - Rifaximin (includes PO) ⁽¹³¹⁾
- Ileus and, **One:** ⁽¹³⁵⁾
 - Surgical intervention required (use **General Surgical subset**)
 - Resolved and initiating PO or tube feeding $\leq 2d$
 - Unresolved and IV fluid $\leq 5d$ since initiation ⁽²²⁾
- Inadequate oral intake and IV fluid $\leq 4d$ ^(22, 136)
- Post Critical care $\leq 24h$
- **INTERMEDIATE, $\geq$ One:**
 - SBP and hepatic encephalopathy Grade III and, $\geq$ **One:** ^(70, 74)
 - Confusion
 - Gross disorientation ⁽⁷⁵⁾
 - Somnolence to semi-stupor and responsive to stimuli ⁽⁷⁶⁾
 - Unusual or bizarre behavior ⁽⁷⁷⁾
 - Hepatorenal syndrome and, $\geq$ **One:** ⁽¹³⁷⁾
 - Albumin
 - Octreotide and midodrine (includes PO)
 - HFNC and respiratory monitoring every 3-4h ^(78, 79)
 - NIPPV ⁽⁸⁰⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ⁽⁸¹⁾
- **CRITICAL, $\geq$ One:**
 - Fulminant *C. difficile* infection and, **Both:** ⁽⁸³⁾
 - Ileus ⁽⁸⁵⁾
 - IV fluid resuscitation $\leq 2d$ ^(84, 86)
 - Severe cholangitis and biliary decompression or drainage $\leq 24h$ ^(87, 92)
 - SBP and hepatic encephalopathy Grade IV ^(74, 93)
 - Toxic megacolon and, $\geq$ **One:** ⁽⁹⁴⁾
 - Blood product transfusion
 - Colonic decompression
 - IV fluid resuscitation $\leq 2d$ ⁽⁸⁶⁾
 - Hepatorenal syndrome and, **Both:** ⁽¹³⁷⁾
 - Hemodynamic instability ⁽¹³⁸⁾
 - Dialysis, **One:**
 - Hemodialysis $\leq 7d$ since initiation
 - Continuous renal replacement therapy (CRRT) ^(97, 139)
 - CRRT ⁽⁹⁷⁾
 - DIC ⁽⁹⁸⁾
 - ECMO or ECLS ⁽¹⁰⁰⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24h$

and, **Both:** ⁽¹⁰¹⁾

● **Hypotension, ≥ One:**

- MAP < 65 mmHg
- Systolic blood pressure < 90 mmHg and < baseline

● **Finding, ≥ One:**

- Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽¹⁰²⁾
- Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽¹⁰³⁾
- HFNC and respiratory monitoring every 1-2h ^(78, 79, 81)
- Mechanical ventilation

● **NIPPV and, ≥ One:** ⁽⁸⁰⁾

- FiO₂ > 50%(0.50)
- Respiratory intervention every 1-2h ⁽¹⁴⁰⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 4-6, One:**

● **ACUTE, One:**

● **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾

- Afebrile
- Laboratory values within acceptable limits ⁽¹¹³⁾
- Tolerating PO ⁽¹¹⁵⁾

● **Infection resolving and, ≥ One:** ⁽¹¹⁷⁾

● **Abscess and, ≥ One:** ⁽¹⁴⁶⁾

- Anti-infective regimen established and tolerated
- Drain removed or drainage manageable
- Acute (ascending) cholangitis and anti-infective regimen established and tolerated ⁽¹¹⁴⁾

● **Appendicitis and, ≥ One:** ^(111, 114)

- Anti-infective regimen established and tolerated
- Drain removed or drainage manageable
- *C. difficile* infection and decrease in frequency and severity of diarrhea
- Diverticulitis and anti-infective regimen established and tolerated ⁽¹¹⁴⁾

● **Pelvic inflammatory disease (includes tubo-ovarian abscess) and, ≥ One:**

- Anti-infective regimen established and tolerated
- Pain controlled or manageable ^(111, 114)
- Drain removed or drainage manageable
- SBP and anti-infective regimen established and tolerated ⁽¹¹⁴⁾
- Toxic megacolon and nutritional route established ⁽¹⁴³⁾

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽¹¹⁸⁾

● **Complication or comorbidity and clinically stable for discharge, ≥ One:**

- Adverse drug reaction resolving
- Blood glucose within acceptable limits ⁽¹¹³⁾

● **Hepatic encephalopathy and, Both:** ⁽⁷⁴⁾

- Grade I or within acceptable limits ⁽¹¹³⁾
- Neurological stability last 24h ⁽¹⁴²⁾

● **Ileus and, One:**

- Tolerating PO ⁽¹¹⁵⁾
- Nutritional route established ⁽¹⁴³⁾

● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**

- Gastrointestinal (GI) Bleeding (**use Gastrointestinal (GI) Bleeding subset**)
- Surgical intervention required (**use General Surgical subset**)
- Abscess, intra-abdominal or pelvic and anti-infective ≤ 6d since initiation
- Acute (ascending) cholangitis and anti-infective ≤ 5d since initiation
- Appendicitis (nonoperative management) and anti-infective ⁽⁴⁰⁾

- Biliary decompression or drainage $\leq 24\text{h}$
- **C. difficile infection and, Both:**
 - Worsening frequency and severity of diarrhea
 - **Intervention, Both:**
 - Anti-infective (includes PO) $\leq 5\text{d}$ since initiation
 - Continuous IV fluid replacement based on losses ⁽¹¹⁶⁾
- Diverticulitis and anti-infective $\leq 5\text{d}$ since initiation
- **Pelvic inflammatory disease and anti-infective $\leq 4\text{d}$ since initiation and, $\geq$ One:**
 - Pregnancy ⁽¹²⁰⁾
 - Severe illness ⁽¹²¹⁾
 - Tubo-ovarian abscess ^(63, 122)
 - **Persistent symptoms, $\geq$ One:**
 - Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$
 - Abdominal pain, persistent
- Spontaneous bacterial peritonitis and anti-infective $\leq 6\text{d}$ since initiation
- Toxic megacolon and parenteral nutrition
- Abscess drainage requiring drain manipulation $\leq 24\text{h}$ ^(123, 124)
- Adverse anti-infective drug reaction requiring change in medication $\leq 24\text{h}$ ⁽¹³²⁾
- **Blood glucose $\leq 24\text{h}$, All:** ⁽¹³³⁾
 - **Finding, $\geq$ One:**
 - $> 250\text{ mg/dL}(13.9\text{ mmol/L})$
 - $< 70\text{ mg/dL}(3.9\text{ mmol/L})$
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(133, 134)
- **Hepatic encephalopathy Grade II and, $\geq$ One:** ^(74, 125, 126)
 - Branched-chain amino acids (BCAA) (includes PO) ⁽¹²⁷⁾
 - Lactulose (includes PO or PR) ⁽¹²⁸⁾
 - L-ornithine L-aspartate (LOLA) (includes PO) ⁽¹²⁹⁾
 - Neomycin (includes PO) ⁽¹³⁰⁾
 - Rifaximin (includes PO) ⁽¹³¹⁾
- **Ileus and, One:** ⁽¹³⁵⁾
 - Surgical intervention required (**use General Surgical subset**)
 - Resolved and initiating PO or tube feeding $\leq 2\text{d}$
 - Unresolved and IV fluid $\leq 5\text{d}$ ⁽²²⁾
- Inadequate oral intake and IV fluid $\leq 4\text{d}$ ^(22, 136)
- Post Critical care $\leq 24\text{h}$
- **INTERMEDIATE, $\geq$ One:**
 - **Spontaneous bacterial peritonitis and hepatic encephalopathy Grade III and, $\geq$ One:** ^(70, 74)
 - Confusion
 - Gross disorientation ⁽⁷⁵⁾
 - Somnolence to semi-stupor and responsive to stimuli ⁽⁷⁶⁾
 - Unusual or bizarre behavior ⁽⁷⁷⁾
 - **Hepatorenal syndrome and, $\geq$ One:** ⁽¹³⁷⁾
 - Albumin
 - Octreotide and midodrine (includes PO)
 - HFNC and respiratory monitoring every 3-4h ^(78, 79)
 - NIPPV ⁽⁸⁰⁾
 - Oxygen $\geq 40\%(0.40)$, $\leq 2\text{d}$ ⁽⁸¹⁾
- **CRITICAL, $\geq$ One:**
 - **Fulminant C. difficile infection and, Both:** ⁽⁸³⁾
 - Ileus ⁽⁸⁵⁾
 - IV fluid resuscitation $\leq 2\text{d}$ ^(84, 86)
 - SBP and hepatic encephalopathy Grade IV ^(74, 93)

- Toxic megacolon and, ≥ **One:** ⁽⁹⁴⁾
 - Blood product transfusion
 - Colonic decompression
 - IV fluid resuscitation ≤ 2d ⁽⁸⁶⁾
- Hepatorenal syndrome and, **Both:** ⁽¹³⁷⁾
 - Hemodynamic instability ⁽¹³⁸⁾
- Dialysis, **One:**
 - Hemodialysis ≤ 7d since initiation
 - Continuous renal replacement therapy (CRRT) ^(97, 139)
- CRRT ⁽⁹⁷⁾
- DIC ⁽⁹⁸⁾
- ECMO or ECLS ⁽¹⁰⁰⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** ⁽¹⁰¹⁾
 - Hypotension, ≥ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Finding, ≥ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽¹⁰²⁾
 - Urine output < 0.5 mL/kg/h
 - Hemodynamic monitoring, invasive ⁽¹⁰³⁾
 - HFNC and respiratory monitoring every 1-2h ^(78, 79, 81)
 - Mechanical ventilation
 - NIPPV and, ≥ **One:** ⁽⁸⁰⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽¹⁴⁰⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 7, **One:**
 - ACUTE, **One:**
 - Responder, discharge expected today if clinically stable last 12h, **All:** ⁽¹¹²⁾
 - Afebrile
 - Laboratory values within acceptable limits ⁽¹¹³⁾
 - Tolerating PO ⁽¹¹⁵⁾
 - Infection resolving and, ≥ **One:** ⁽¹¹⁷⁾
 - Abscess and, ≥ **One:** ⁽¹⁴⁶⁾
 - Anti-infective regimen established and tolerated
 - Drain removed or drainage manageable
 - Acute (ascending) cholangitis and anti-infective regimen established and tolerated ⁽¹¹⁴⁾
 - Appendicitis and, ≥ **One:** ^(111, 114)
 - Anti-infective regimen established and tolerated
 - Drain removed or drainage manageable
 - *C. difficile* infection and decrease in frequency and severity of diarrhea
 - Diverticulitis and anti-infective regimen established and tolerated ⁽¹¹⁴⁾
 - Spontaneous bacterial peritonitis and anti-infective regimen established and tolerated ⁽¹¹⁴⁾
 - Toxic megacolon and nutritional route established ⁽¹⁴³⁾
 - Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽¹¹⁸⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
 - Adverse drug reaction resolving
 - Blood glucose within acceptable limits ⁽¹¹³⁾
 - Hepatic encephalopathy and, **Both:** ⁽⁷⁴⁾
 - Grade I or within acceptable limits ⁽¹¹³⁾
 - Neurological stability last 24h ⁽¹⁴²⁾

- **Ileus and, One:**
 - Tolerating PO ⁽¹¹⁵⁾
 - Nutritional route established ⁽¹⁴³⁾
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - Use Extended Stay subset
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - Use Extended Stay subset
 - Refer for secondary review
- **CRITICAL, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - Use Extended Stay subset
 - Refer for secondary review

Discharge, Level of Care, $\geq$ One: ⁽¹⁴⁷⁾

- **HOME, Both:**
 - Level of care appropriateness, **Both:**
 - Home environment safe and accessible ⁽¹⁴⁸⁾
 - Patient or caregiver demonstrates ability to manage care
 - Gastrointestinal or gynecological infection discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽¹⁴⁹⁾
 - Medication reconciliation ⁽¹⁵⁰⁾
 - Follow-up care planned ⁽¹⁵¹⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽¹⁴⁸⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(152, 153)
 - Skilled services, $\geq$ **One:** ⁽¹⁵⁴⁾
 - Skilled nursing ⁽¹⁵⁵⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽¹⁵⁶⁾
 - Gastrointestinal or gynecological infection discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽¹⁴⁹⁾
 - Medication reconciliation ⁽¹⁵⁰⁾
 - Follow-up care planned ⁽¹⁵¹⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽¹⁵⁷⁾
 - Skilled treatment, $\geq$ **One:**
 - Clinical assessment ⁽¹⁵⁸⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽¹⁵⁹⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq$ 1x/wk
 - Treatment precluded in a lower level

- **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽¹⁵⁵⁾
- **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(160, 161)
 - Functional impairments requiring at least supervision ⁽¹⁶²⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽¹⁵⁰⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - **Services, ≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽¹⁵⁵⁾
 - **Therapy, All:**
 - Goal directed therapy with rehabilitation potential ^(160, 161)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁶²⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽¹⁵⁰⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - **Level of care appropriateness, All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(160, 161)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁶²⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽¹⁵⁰⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - **Level of care appropriateness, Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - **Services, All:**
 - Goal directed therapy with rehabilitation potential ^(160, 161)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽¹⁶²⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽¹⁶³⁾
 - **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽¹⁵⁰⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - **Level of care appropriateness, Both:**

- Treatment precluded in a lower level
- In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One**:
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽¹⁶⁴⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(165, 166)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽¹⁶⁷⁾
- Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽¹⁵⁰⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
371 MAJOR GASTROINTESTINAL DISORDERS AND PERITONEAL INFECTIONS WITH MCC	n/a	5.4	CMS GMLOS
372 MAJOR GASTROINTESTINAL DISORDERS AND PERITONEAL INFECTIONS WITH CC	n/a	3.8	CMS GMLOS
373 MAJOR GASTROINTESTINAL DISORDERS AND PERITONEAL INFECTIONS WITHOUT CC/MCC	n/a	2.8	CMS GMLOS
391 ESOPHAGITIS, GASTROENTERITIS AND MISCELLANEOUS DIGESTIVE DISORDERS WITH MCC	n/a	3.9	CMS GMLOS
392 ESOPHAGITIS, GASTROENTERITIS AND MISCELLANEOUS DIGESTIVE DISORDERS WITHOUT MCC	n/a	2.6	CMS GMLOS
444 DISORDERS OF THE BILIARY TRACT WITH MCC	n/a	4.4	CMS GMLOS
445 DISORDERS OF THE BILIARY TRACT WITH CC	n/a	3.1	CMS GMLOS
446 DISORDERS OF THE BILIARY TRACT WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
757 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITH MCC	n/a	5	CMS GMLOS
758 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITH CC	n/a	3.7	CMS GMLOS
759 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITHOUT CC/MCC	n/a	2.6	CMS GMLOS
867 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH MCC	n/a	5.5	CMS GMLOS
868 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH CC	n/a	3.7	CMS GMLOS
869 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
APPENDICITIS	56%	1.6	InterQual

Condition or Procedure	% Pd Obs	LOS (days)	Type
ASCENDING CHOLANGITIS	22%	3.9	InterQual
C. DIFFICILE COLITIS	17%	4.9	InterQual
DIVERTICULITIS	17%	3.8	InterQual
INTRA-ABDOMINAL ABSCESS	9%	5.7	InterQual
PELVIC ABSCESS	36%	3.3	InterQual
PELVIC INFLAMMATORY DISEASE	36%	3.1	InterQual
SPONTANEOUS BACTERIAL PERITONITIS	6%	5	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

These criteria apply to patients with suspected acute appendicitis in whom imaging is non-diagnostic or equivocal, and who need additional evaluation to rule in or out the diagnosis of acute appendicitis.

Instruction: If a patient requires surgical intervention, refer to the General Surgical subset.

5:

Pain associated with acute appendicitis varies according to the anatomic location of the appendix; a pelvic appendix can cause tenderness in the suprapubic area or the right lower quadrant, while a retrocecal appendix may manifest as periumbilical or flank discomfort (Herr et al., Pain Manag Nurs 2019, 20: 404-17).

6:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

7:

Patients with mild to moderate active vomiting should be treated with intravenous hydration and antiemetic therapy. A dehydrated patient who cannot tolerate oral hydration should not be discharged (Joshua Guttman, In: Rosen's Emergency Medicine, 9 edn. 2018:230-41.e1).

8:

Elevation of white blood cell count (WBC) greater than 10,000/cu.mm($10 \times 10^9/L$) is present in 67%(0.67)-90%(0.90) of patients presenting with acute appendicitis. Mild leukocytosis and abdominal pain are signs of acute appendicitis and will require further workup for a definitive diagnosis (Moris et al., Jama 2021, 326: 2299-311).

9:

Computed tomography (CT) has a high diagnostic accuracy with a sensitivity and specificity of 86% and 95%, respectively, in adults undergoing evaluation for suspected appendicitis (Garcia et al., J Am Coll Radiol 2018, 15: S373-s87). While CT has a relatively short acquisition time, it exposes the patient to ionizing radiation, and may not always be appropriate (e.g., children, pregnant patients) (Repplinger et al., Radiology 2018, 288: 467-75). With sensitivity and specificity of 96% and 97%, respectively, Magnetic resonance imaging (MRI) is an alternative to CT in the evaluation of suspected appendicitis due to its lack of ionizing radiation but may not be appropriate for patients with contraindications (e.g., metal implants, claustrophobia) (Repplinger et al., Radiology 2018, 288: 467-75; Duke et al., AJR Am J Roentgenol 2016, 206: 508-17). A meta-analysis of 19 studies showed MRI sensitivity and specificity to be 92% and 98% in the pregnant patient with abdominal pain and suspected appendicitis (Kave et al., World J Emerg Surg 2019, 14: 37). Ultrasound (US) is inexpensive, convenient, and avoids the risks associated with radiation and contrast. However, unlike CT and MRI, patient's age and body habitus influence its diagnostic accuracy. US has been shown to be more sensitive and specific in children than in adults with acute appendicitis (Garcia et al., J Am Coll Radiol 2018, 15: S373-s87; Kabir et al., Int J Surg 2017, 40: 155-62; Cundy et al., J Pediatr Surg 2016, 51: 1939-43). The sensitivity of US can be enhanced by repeating studies in patients with persistent symptoms if the first study is negative or equivocal, since evolving appendiceal inflammation may become more readily detectable on a subsequent ultrasound after a period of observation (Eng et al., Radiology 2018, 288: 717-27; Kim et

al., Br J Radiol 2018, 91: 20170529; Schuh et al., Acad Emerg Med 2015, 22: 406-14).

10:

Imaging includes:

- Ultrasound (US)
- Computed tomography (CT)
- Magnetic resonance imaging (MRI)

11:

Patients with appendicitis diagnosed by ultrasound (US) may require additional imaging studies to guide further management. Computed tomography (CT) and magnetic resonance imaging (MRI) may be utilized to detect complications such as abscess formation or bowel obstruction that may not be detected or fully characterized by US. CT accuracy is not hampered by surgical wounds, drains, or dressings. MRI has excellent soft tissue resolution, can be used for diagnostic clarification of findings involving the inflammation and anatomical distortion associated with complex appendicitis, and can help differentiate comorbid pelvic conditions; but is not always as readily available as CT (Mervak et al., J Magn Reson Imaging 2019, 50: 1367-76; Eng et al., Radiology 2018, 288: 717-27; Kim et al., Br J Radiol 2018, 91: 20170529; Schuh et al., Acad Emerg Med 2015, 22: 406-14).

12:

Clostridioides difficile testing may be positive in patients without symptoms who have active colonization, therefore only patients with symptoms should be tested. These symptoms include 3 or more loose stools within 24 hours or the presence of pseudomembranous colitis identified by imaging or colonoscopy. Guidelines recommend the use of a multistep algorithm combining more than one type of testing for the most accurate results (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44; Poylin et al., Dis Colon Rectum 2021, 64: 650-68).

13:

American College of Gastroenterology classifies the severity of *Clostridioides difficile* infections as follows (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47):

- Mild disease - diarrhea only
- Moderate disease - diarrhea and additional symptoms that do not meet the acuity of severe disease
- Severe disease - increased white blood cell (WBC) count or increased serum creatinine
- Fulminant disease - meeting criteria for severe *C. difficile* infection combined with hypotension/shock, ileus, or megacolon

14:

Patients experiencing continued deterioration of their infection demonstrate symptoms which may include fever, pain, mental status change, hypotension, hypoxia, nausea, and/or vomiting.

15:

A patient should demonstrate objective signs of clinical improvement within 72 hours of initiation of treatment.

16:

Instruction: This criteria point does not include failed treatment at the Observation level of care.

These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic, trial of home care services, or emergency department. These criteria can be applied when the patient fails to respond to more intensive treatment beyond the routine treatments or medications delivered for a chronic condition.

17:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with

outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

18:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

19:

Tachycardia and orthostatic hypotension are common symptoms of dehydration when vomiting or diarrhea are present (Tomas Nguyen and Saadia Akhtar, In: Rosen's Emergency Medicine, 9 edn. 2018:1129-49.e2).

20:

Orthostatic hypotension (Juraschek et al., JAMA Intern Med 2017, 177: 1316-23).

21:

Hypernatremia and dehydration (Camiron L. Pfennig and Corey M. Slovis, In: Rosen's Emergency Medicine. 2018: 1516-32.e2)

22:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

23:

Some patients who present with uncomplicated diverticulitis can be managed outpatient with bowel rest and diet modifications (e.g., fluids, clear liquid). Treatment with parenteral antibiotics is still recommended for patients who have significant comorbidities, signs of systemic inflammatory response, or are at risk for progressing to complicated diverticulitis (Qaseem et al., Ann Intern Med 2022, 175: 399-415; Hall et al., Diseases of the Colon & Rectum 2020, 63: 728-47; Sartelli et al., World J Emerg Surg 2020, 15: 32).

24:

Instruction: Uncomplicated diverticulitis is confirmed when the computed tomography (CT) shows localized inflammation or thickening of the colon wall.

25:

Computed tomography (CT) is the preferred imaging for the diagnosis of diverticulitis, as it is highly sensitive and specific. CT is recommended when a patient has failed outpatient therapy and needs to be evaluated further for possible complications, such as an abscess or perforation (Peery et al., Gastroenterology 2021, 160: 906-11.e1; Sartelli et al., World J Emerg Surg 2020, 15: 32).

26:

Although some patients with uncomplicated diverticulitis can be easily treated in the outpatient setting and without antibiotics, guidelines recommend that patients with risk factors for progression to complicated diverticulitis receive parenteral antibiotics (Qaseem et al., Ann Intern Med 2022, 175: 399-415; Peery et al., Gastroenterology 2021, 160: 906-11.e1):

- Severe comorbid conditions or frailty
- Increased CRP > 140 mg/L
- A pain score of > 7 defines severe pain upon presentation

- Vomiting
- Increased WBC

27:

Severe systemic disorders classified as III or IV by The American Society of Anesthesiologists (ASA) include (but are not limited to):

Class III

- Substantial functional limitations
- Poorly controlled DM
- Poorly controlled HTN
- COPD
- Morbid obesity (BMI ≥ 40 kg/m²)
- Active hepatitis
- Alcohol dependence or abuse
- Implanted pacemaker
- Moderate reduction of ejection fraction
- ESRD and dialysis
- History (>3 months) of MI, CVA, TIA, or CAD/stents

Class IV

- Recent (<3 months) MI, CVA, TIA or CAD/stents
- Ongoing cardiac ischemia
- Severe valve dysfunction
- Severe reduction of ejection fraction
- ESRD not requiring dialysis

28:

The guidelines recommend that patients presenting with acute uncomplicated diverticulitis who have features of severe chronic illness as classified by The American Society of Anesthesiologists (ASA) classification III or IV, have a 5% higher risk of progression from an uncomplicated to a complicated diverticular process (Peery et al., Gastroenterology 2021, 160: 906-11.e1; American Society of Anesthesiologists, ASA Physical Status Classification System. 2020).

29:

A pain score of greater than 7 upon presentation may be indicative of complicated diverticulitis or may indicate progression from uncomplicated to complicated disease (Qaseem et al., Ann Intern Med 2022, 175: 399-415).

30:

Symptoms of acute diverticulitis that last longer than 5 days prior to hospitalization warrant additional imaging and monitoring for possible progression to complicated diverticulitis. These symptoms may include, but are not limited to (Peery et al., Gastroenterology 2021, 160: 906-11.e1):

- Abdominal pain (tenderness or cramping)
- Fever
- Change in bowel habits (constipation)
- Nausea without vomiting or anorexia

31:

Symptoms of possible progression to complicated diverticulitis may include:

- Abdominal pain (tenderness or cramping)
- Fever
- Change in bowel habits (constipation)
- Nausea without vomiting or anorexia

32:

Systemic inflammatory response syndrome (SIRS) is an inflammatory process defined as 2 or more of the following

clinical findings: fever, leukocytosis, tachypnea, and tachycardia.

33:

Pelvic inflammatory disease (PID) is a term that encompasses several pelvic inflammatory processes, usually of infectious etiology, including endometritis, salpingitis, tubo-ovarian abscess, and pelvic peritonitis.

Presenting symptoms may include:

- Abdominal pain
- Fever or chills
- Abnormal vaginal discharge
- Vomiting

A diagnosis can be made clinically after a bimanual exam that is positive for cervical motion tenderness, uterine or adnexal pain, or adnexal mass. If abnormal vaginal discharge is present, a speculum exam may be warranted (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187; Curry et al., Am Fam Physician 2019, 100: 357-64). Pelvic ultrasound is useful to rule out other causes of pelvic pain, such as appendicitis (Dewey and Wittrock, Emerg Med Clin North Am 2019, 37: 207-18).

34:

Oral anti-infectives that cause side effects may not treat the underlying infection appropriately. Patients with anti-infective side effects may not adhere to scheduling of the medication appropriately, may stop taking the medication or may present to the emergency room for symptom management. Nausea and vomiting may preclude tolerance or completion of the anti-infective and this may require IV anti-infectives and monitoring (Hale et al., J Pharm Pract 2017, 30: 600-5).

Side effects of oral anti-infectives may include:

- Diarrhea
- Nausea/vomiting
- Abdominal pain/cramping
- Rash or hives
- Angioedema

35:

Patients with suspected pelvic inflammatory disease (PID) need further workup to accurately diagnose the condition. PID symptoms can imitate other conditions, such as ectopic pregnancy, acute appendicitis, ovarian cyst, or ovarian torsion that may need emergency surgical intervention (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187).

36:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

37:

Patients with an intra-abdominal or a pelvic abscess typically present with a sudden onset of abdominal pain along with gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea). A fever may or may not be present. Computed tomography (CT) is the preferred diagnostic imaging technique due to high accuracy. The size and location of the abscess will determine whether treatment with intravenous anti-infectives alone is sufficient (e.g., smaller fluid collections) and/or when drainage may be appropriate. If symptoms worsen or do not improve with antibiotics alone, drainage may subsequently be needed (DeCesare et al., Am Surg 2021, 87: 120-4; Thompson and Hrabe, Gastroenterol Clin North Am 2021, 50: 475-88; Workowski et al., MMWR Recomm Rep 2021, 70: 1-187).

38:

Imaging is necessary for establishing a diagnosis of a pelvic or intra-abdominal abscess. Ultrasound (US) if available, may be performed first, although computed tomography (CT) is the best imaging modality (Cao et al.,

Gynecol Oncol 2021, 162: 107-12; Olpin and Strachowski, Radiol Clin North Am 2020, 58: 329-45).

39:

Imaging for pelvic or intra-abdominal abscess may include:

- Computed tomography (CT)
- Ultrasound (US)

40:

Complicated appendicitis with an abscess or phlegmon identified on imaging may be managed non-operatively with antibiotics, drainage, or both, with a possible interval (delayed) appendectomy several weeks later. Some abscesses cannot be drained percutaneously and complicated appendicitis treated nonoperatively can fail to respond to treatment or recur, the role of surgery during the initial admission (emergency appendectomy, immediate appendectomy, or early appendectomy) may be required (Di Saverio et al., World J Emerg Surg 2020, 15: 27; Gavriilidis et al., J Clin Med Res 2019, 11: 56-64; Garcia et al., J Am Coll Radiol 2018, 15: S373-s87).

41:

Complicated appendicitis is when there is an uncontained rupture or abscess of phlegmon (contained rupture).

42:

Abdominal imaging for appendicitis may include:

- Computed tomography (CT)
- Magnetic resonance imaging (MRI)
- Ultrasound (US)

43:

The recommended treatment for complicated appendicitis with an appendiceal abscess or phlegmon is prompt initiation of anti-infectives and percutaneous drainage if amenable. If percutaneous drainage is not feasible or appropriate, then anti-infectives should be given and surgery should be considered (Nimmagadda et al., Am J Surg 2019, 217: 713-7).

44:

The classic signs and symptoms of cholangitis are abdominal pain, jaundice, and fever (Charcot's Triad). Patients do not always present with the classic symptoms and in many cases may present with one or a combination of symptoms or findings below (An et al., Gastroenterol Clin North Am 2021, 50: 403-14):

- Abdominal pain
- Fever
- Jaundice
- Increased bilirubin
- Shaking chills
- Increased or decreased WBC
- Increased CRP

Patients of advanced age (80 years or older) may not present with the classic symptoms of acute cholangitis (fever, abdominal pain, jaundice). They may have vague symptoms of generalized weakness and malaise, often accompanied by laboratory evidence of cholestasis (Thuluvath et al., Mayo Clin Proc Innov Qual Outcomes 2021, 5: 377-87).

In patients with mild disease, biliary decompression should be performed within 24 to 48 hours after stabilization (Buxbaum et al., Gastrointest Endosc 2021, 94: 207-21.e14).

45:

Instruction: Patients with severe or progressive disease require urgent biliary drainage and management of organ dysfunction and should be reviewed at the Critical level of care.

46:

Patients of advanced age (80 years or older) may not present with the classic symptoms of acute cholangitis (fever, abdominal pain, jaundice). They may have vague symptoms of generalized weakness and malaise, often accompanied by laboratory evidence of cholestasis (Thuluvath et al., Mayo Clin Proc Innov Qual Outcomes 2021, 5: 377-87).

47:

Procalcitonin (PCT) is a biomarker that can be used to identify bacterial infections or sepsis. PCT may be utilized as an alternative lab value to CRP as it can sometimes help to differentiate a bacterial versus a viral infection. PCT rises earlier and normalizes faster compared with the trajectory for CRP (Samsudin and Vasikaran, Clin Biochem Rev 2017, 38: 59-68).

48:

Ultrasound (US) is the preferred initial imaging for suspected cholangitis. US is highly specific (94-100%) for identifying bile duct stones or dilation when properly performed. If US is unavailable, inconclusive, or not able to be performed (e.g., morbid obesity), abdominal computed tomography (CT), magnetic resonance cholangiopancreatography (MRCP) or endoscopic ultrasound (EUS) are feasible alternatives (An et al., Gastroenterol Clin North Am 2021, 50: 403-14; Miura et al., J Hepatobiliary Pancreat Sci 2018, 25: 31-40).

49:

Imaging studies include:

- Ultrasound (US)
- Computed tomography (CT)
- Magnetic resonance cholangiopancreatography (MRCP)
- Endoscopic ultrasound (EUS)

Imaging findings indicative of cholangitis may include:

- Biliary dilation
- Biliary stone, stricture, or stent
- Obstructing mass

50:

The anti-infective regimen for cholangitis can range between 4 to 7 days after source control is achieved (Miura et al., J Hepatobiliary Pancreat Sci 2018, 25: 31-40).

51:

Patients presenting with *Clostridioides difficile* infection associated with an increased WBC greater than 15,000 cells/mm³ or a serum creatinine greater than 1.5 are at risk for poor outcomes and are classified as severe per the guidelines. Severe *Clostridioides difficile* classification also includes serum creatinine greater or equal to 1.5x baseline or estimated baseline to account for the patient with chronic renal insufficiency (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47; Cohen et al., Infect Control Hosp Epidemiol 2010; 31(5): 431-455).

52:

A temperature greater than 101.3°F(38.5°C) is an unlikely finding of severe *C. difficile* infection, although, when present it is an effective predictor of a poor outcome (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47).

53:

When the baseline serum creatinine level is unknown, the Modification of Diet in Renal Disease (MDRD) equation can be used to estimate an expected creatinine value (assuming a normal baseline value for glomerular filtration rate (GFR) of 75ml/min per 1.73 meters squared). The following values demonstrate one method to estimate baseline serum creatinine (eScr) (with the omission of the race variable) (Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138; Bellomo et al., Crit Care 2004, 8: R204-12):

Male sex assigned at birth:

- Age 20-24 - eScr 1.3 mg/dL (115 µmol/l)

- Age 25-39 - eSCr 1.2 mg/dL (106 µmol/l)
- Age 40-65 - eSCr 1.1 mg/dL (97 µmol/l)
- Age > 65 - eSCr 1.0 mg/dL (88 µmol/l)

Female sex assigned at birth:

- Age 20-29 - eSCr 1.0 mg/dL (88 µmol/l)
- Age 30-54 - eSCr 0.9 mg/dL (80 µmol/l)
- Age > 54 - eSCr 0.8 mg/dL (71 µmol/l)

Both the MDRD and the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations provide a calculation of estimated GFR using serum creatinine, sex assigned at birth, age, and race. Recognition of substantial diversity within population groups has led to multiple organizations no longer including race in the estimated GFR calculations. This approach was investigated by a joint task force of the National Kidney Foundation and the American Society of Nephrology, the result of which was a recommendation to implement the new CKD-EPI creatinine equation which does not use the race variable (Delgado et al., Am J Kidney Dis 2022, 79: 268-88 e1; Inker et al., N Engl J Med 2021, 385: 1737-49).

54:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

55:

Computed tomography (CT) findings of complicated diverticulitis may include:

- Perforation
- Microperforation
- Abscess
- Fistula
- Ileus
- Obstruction
- Peritonitis

56:

Computed tomography (CT) findings of complicated diverticulitis may include (Sartelli et al., World J Emerg Surg 2020, 15: 32):

- Perforation
- Microperforation
- Abscess
- Fistula
- Ileus
- Obstruction
- Peritonitis

57:

Although there is no consensus on standard treatment for diverticular abscesses, a combination of percutaneous drainage (when needed) and antibiotics have demonstrated favorable outcomes (Sartelli et al., World J Emerg Surg 2020, 15: 32).

Treatment based on specific size of diverticular abscess or pericolic gas may include (van Dijk et al., Surg Infect (Larchmt) 2018, 19: 362-8):

- Large abscesses (3 - 5cm): may require drainage and anti-infectives
- Small abscesses (< 3cm; stable patients): anti-infective alone can be effective
- Pericolic extraluminal gas: anti-infective alone can be effective

58:

The treatment recommendation for diverticulitis with microperforation (extraluminal air, localized perforation) is limited within the literature. Because there are few studies and small sample sizes, a clear treatment approach has not been developed. At this time, nonoperative management with anti-infectives is an appropriate treatment plan

and has shown to be effective with a 94% success rate (Hall et al., Dis Colon Rectum 2020, 63: 728-47; van Dijk et al., Surg Infect (Larchmt) 2018, 19: 362-8).

59:

Patients who are immunocompromised (e.g., end-stage liver disease, end-stage renal disease, other immunocompromising conditions, or immunosuppressing therapy) are at a higher rates of requiring emergency surgery and high rate of recurrence after initial episode being treated medically (Hall et al., Dis Colon Rectum 2020, 63: 728-47).

60:

Immunocompromised refers to the patient who has a decreased capacity to defend against invading organisms as a result of an impaired immune system due to several comorbidities. The principal manifestation of immunodeficiency is increased susceptibility to infection with unusual severity, frequency, and duration which differs from that of an individual who is immunocompetent who may be exposed to the same pathogen. Furthermore, individuals who are immunocompromised are increasingly prone to opportunistic infections. Immunodeficiency is classified as either primary (congenital), caused by inherited disorders of the immune system, or secondary (acquired), as a result of a disease process or as a result of treatment of other chronic diseases. Primary causes of immunodeficiency account for more than 200 genetic disorders that impact immune system function. Some examples of primary immunodeficiency include congenital immunodeficiency diseases (e.g., X-linked agammaglobulinemia), combined immunodeficiency disease (e.g., DiGeorge syndrome, Wiskott syndrome), and chronic granulomatous disease.

Examples of secondary immune deficiency include poorly controlled diabetes mellitus defined as HbA1c 9%(0.09) (drawn on admission or within the past three months), end-stage liver disease, hematopoietic malignancies, patients who have had allogeneic hematopoietic stem cell transplantation (HSCT), and patients receiving immunosuppressive therapy (e.g., chemotherapy, radiation, anti-rejection medications, or prolonged high steroid use). Patients with severe, prolonged neutropenia (e.g., absolute neutrophil count less than 500/cu.mm(500x106/L), HSCT, human immunodeficiency virus (HIV) with CD4 count less than 200/cu.mm(200x106/L), a history of splenectomy, splenic dysfunction due to sickle cell disease, and those who have received intense chemotherapy (e.g., childhood acute myelogenous leukemia) are at greatest risk of infection (Centers for Disease Control and Prevention, National Diabetes Statistics Report. 2022; Bonilla et al., J Allergy Clin Immunol 2015, 136: 1186-205 e78; Lipska et al., Diabetes Care 2013, 36: 3535-42).

61:

The following categories of organ dysfunction are derived from the Sequential [Sepsis-Related] Organ Failure Assessment (SOFA) score, as proposed in the Third International Consensus Definition for Sepsis and Septic Shock (Sepsis-3), with the following modifications: acute kidney injury is defined as an increase in serum creatinine with a reference to baseline creatinine (and not absolute numeric values used in SOFA), and oxygen saturation is used in place of the PaO2/FiO2 (partial pressure of oxygen in arterial blood/fraction of inspired oxygen) ratio given the lack of universal availability of arterial blood gas measurements for patients at the Acute level of care (Singer et al., JAMA 2016, 315: 801-10; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

62:

Several studies have shown that patients with moderately elevated lactic acid levels (greater than normal but under 4 mmol/L) and evidence of infection are at increased risk for poor outcome (Villar et al., Infect Dis (Auckl) 2019, 12: 1178633719862776; Oedorf et al., West J Emerg Med 2017, 18: 258-66; Rhee et al., JAMA 2017, 318: 1241-9; Puskarich et al., J Crit Care 2014, 29: 334-9; Shapiro et al., Ann Emerg Med 2005, 45: 524-8).

63:

Instruction: Abscess drainage can be percutaneous, ultrasound-guided, or computed tomography-guided. If surgical drainage is required use the General Surgical subset.

64:

According to the Centers for Disease Control and Prevention, patients with severe disease or a high risk of developing complications require admission for intravenous anti-infective.

These may include (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187):

- Pregnancy
- Presence of a tubo-ovarian abscess
- Fever > 101°F
- Vomiting
- Failed outpatient treatment

65:

Pelvic inflammatory disease (PID) during pregnancy places the mother at increased risk for complications of premature delivery and maternal morbidity. Due to these risk factors, pregnant patients should be admitted for anti-infectives, infectious disease consult and close monitoring (Curry et al., Am Fam Physician 2019, 100: 357-64).

66:

Fever greater than 101°F (38.3°C) may be an indication of severe illness in patients with pelvic inflammatory disease (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187).

67:

The preferred treatment of tubo-ovarian abscess is intravenous broad-spectrum anti-infectives and in some cases drainage. If medical management alone is not sufficient and symptoms persist or worsen (within 72 hours) then abscess drainage (ultrasound-guided or computed tomography-guided drainage) is required (if surgical drainage is required use General Surgical subset) (Goje et al., J Minim Invasive Gynecol 2021, 28: 556-64; Weiss et al., J Am Coll Radiol 2020, 17: S265-s80).

68:

Diagnosis of tubo-ovarian abscess (TOA) can be made by several modes of imaging:

- Ultrasound (US)
- Computed tomography (CT)
- Scintigraphy

69:

A patient who is unresponsive to outpatient anti-infectives may have vomiting, persistent pain or a low-grade fever. Patients presenting with no improvement or worsening signs and symptoms despite outpatient antibiotic treatment require admission for IV anti-infectives and further evaluation (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187).

70:

Spontaneous bacterial peritonitis (SBP) is the abrupt onset of bacterial peritonitis in the absence of a recognizable cause of secondary peritonitis (e.g., perforation). It is diagnosed when ascitic fluid polymorphonuclear leukocyte (PMN) count is greater than 250 c.mm($25 \times 10^9/L$). Patients may present with signs of fever, abdominal pain, nausea, encephalopathy, and deterioration in renal function. Treatment for SBP includes intravenous antimicrobials and volume expansion with albumin (Biggins et al., Hepatology 2021, 74: 1014-48; Mattos et al., Ann Hepatol 2020, 19: 451-7; Devani et al., J Clin Gastroenterol 2019, 53: e68-e74).

71:

Albumin administration in patients with spontaneous bacterial peritonitis (SBP) is recommended because it not only aids with intravascular volume, it also improves survival by inhibiting the progression of acute kidney injury (AKI). Anti-infectives combined with albumin have been shown to decrease mortality from 29% to 10% (Biggins et al., Hepatology 2021, 74: 1014-48; Niu et al., Dig Dis Sci 2018, 63: 1327-33; Sort et al., N Engl J Med 1999, 341: 403-9).

72:

Patients with spontaneous bacterial peritonitis (SBP) should receive a third-generation cephalosporin (IV) as first line treatment only when a multi-drug resistant organism (MDRO) is not likely the cause of the infection. A broader anti-infective should be considered in patients who have been recently discharged from a facility (Biggins et al., Hepatology 2021, 74: 1014-48; Marciano et al., Hepat Med 2019, 11: 13-22).

73:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

74:

Hepatic encephalopathy (HE) is a complication of acute or chronic liver disease that is primarily characterized by changes in personality, consciousness, cognition, and motor function. West Haven criteria for grading HE includes (Montrief et al., Am J Emerg Med 2019, 37: 329-37; Weissenborn, Drugs 2019, 79: 5-9; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47):

- Grade I - Lack of awareness, anxiety, short attention span, euphoria, inability to add/subtract
- Grade II - Disorientation for time, inappropriate behavior, lethargy/apathy, obvious personality change, dyspraxia, or asterixis
- Grade III - Confusion, bizarre behavior, gross disorientation, responsive to verbal stimuli, somnolence to semi-stupor
- Grade IV - Coma

75:

Gross disorientation is defined as being disoriented to time and space (Biggins et al., Hepatology 2021, 74: 1014-48; Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47).

Disorientation to time includes at least 3 of the following:

- Disoriented to the date of the month
- Disoriented to the day of the week
- Unable to discern the month or season
- Disoriented to the year

Disorientation to space includes all of the following:

- Unable to identify the country
- Unable to identify the city, state, or region
- Disoriented to place

76:

Grade III West Haven Criteria is limited by the subjectivity of the practitioner's clinical assessment of the patient's signs and symptoms. To satisfy this criterion point patients should be able to respond to verbal stimuli (Dellatore et al., Clin Liver Dis 2020, 24: 189-96; Ferenci et al., Hepatology 2002, 35: 716-21).

77:

West Haven criteria for hepatic encephalopathy (HE) includes the criterion "bizarre behavior" for Grade III HE. This behavior may be viewed as "out of sync" from societal norms; it varies from patient to patient and is often communicated to a healthcare provider by a patient's family member or caregiver (Ferenci et al., Hepatology 2002, 35: 716-21).

78:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO₂) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO₂. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to

standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

79:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

80:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

81:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

82:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

83:

A *Clostridioides difficile* infection (CDI) is considered fulminant (severe) when associated with any one of the following (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47):

- Hypotension/shock
- Ileus
- Megacolon

Instruction: Patients who are diagnosed with fulminant CDI and are hypotensive/exhibiting symptoms of shock or have a confirmed megacolon will meet at the Critical level of care in this subset.

84:

Within the first 72 hours, volume resuscitation and vancomycin (oral or rectal) are the initial treatments of fulminant *Clostridioides difficile* infection. In addition to vancomycin (oral or rectal), intravenous metronidazole should be added when an ileus is present (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47).

85:

Due to the lack of normal peristalsis in the intestine, an ileus can slow down the absorption of oral vancomycin. Vancomycin delivered via enema is an alternative route that may offer greater benefit of absorption (Kelly et al., Am J Gastroenterol 2021, 116: 1124-47; McDonald et al., Clin Infect Dis 2018).

86:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

87:

The clinical presentation of severe cholangitis includes evidence of organ dysfunction. Patients with severe cholangitis should be managed at the Critical level of care and often require fluid resuscitation and urgent biliary drainage which may be performed endoscopically, percutaneously, or surgically (if surgical procedure required use the General Surgical subset) (An et al., Gastroenterol Clin North Am 2021, 50: 403-14; Navuluri et al., Semin Intervent Radiol 2020, 37: 14-23).

88:

When acute cholangitis is complicated by organ dysfunction, it is characterized as severe and biliary decompression or drainage should be performed as soon as possible. Severe cholangitis has a high incidence of bacteremia, and can progress rapidly causing hemodynamic instability (Miura et al., J Hepatobiliary Pancreat Sci 2018, 25: 31-40).

Signs and symptoms of severe cholangitis include (An et al., Gastroenterol Clin North Am 2021, 50: 403-14):

- Change in mental status/new disorientation
- Coagulation dysregulation (increased INR)
- Impaired respiratory function/ respiratory failure
- Thrombocytopenia/ internal bleeding
- Kidney failure

89:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

90:

For patients without anticoagulation therapy or underlying coagulopathy, the international normalized ratio (INR) should be 1.1 or less.

91:

Platelet transfusion may be considered when the following conditions are present (Rhodes et al., Crit Care Med 2017, 45: 486-552):

- When the patient's platelet count is 10,000/cu.mm($10 \times 10^9/L$) or less without the presence of bleeding
- When the patient's platelet count is 20,000/cu.mm($20 \times 10^9/L$) or less and the patient is at risk for bleeding (e.g., temperature less than 100.4°F(38°C), recent hemorrhage, rapid decrease in platelet count, or other coagulation abnormalities)
- When the patient's platelet count is low in the presence of bleeding

92:

Endoscopic retrograde cholangiopancreatography (ERCP) is the recommended procedure for biliary decompression. When ERCP cannot be performed, percutaneous drainage should be considered. ERCP within 48 hours significantly reduces inpatient mortality and is associated with a shorter length of stay (An et al., Gastroenterol Clin North Am 2021, 50: 403-14; Buxbaum et al., Gastrointest Endosc 2021, 94: 207-21.e14).

93:

Coma is classified as Grade IV hepatic encephalopathy according to the West Haven criteria.

94:

Toxic megacolon is defined as colonic dilation (greater than 6 cm) with three of the following symptoms:

- Fever
- Leukocytosis
- Tachycardia
- Anemia/bleeding

In addition to the above symptoms, the patient must have one of the following:

- Hypotension
- Hypovolemia
- Mental status change
- Electrolyte imbalance

Since hypokalemia and hypomagnesemia can exacerbate the dilation, these electrolyte imbalances should be aggressively corrected when present. Patients with toxic megacolon should be treated with intravenous hydration, broad-spectrum antibiotics, and bowel rest. Nasogastric decompression may be considered to remove air and fluid from the colon. Rapid deterioration, continued toxicity, or lack of improvement may be an indication for prompt colectomy. In most cases, surgery is performed after a short trial of medical therapy (Desai et al., Clin Exp Gastroenterol 2020, 13: 203-10; Gajendran et al., Disease-a-Month 2019:).

95:

An abdominal x-ray revealing colonic dilatation of more than 6 cm, without obstruction, is one of the cardinal signs of toxic megacolon. Abdominal computed tomography (CT) may also be used to confirm the diagnosis (Mounsey et al., Am Fam Physician 2020, 101: 168-75).

96:

Toxic megacolon imaging includes:

- Abdominal x-ray
- Abdominal computed tomography (CT)

97:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

98:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as

sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

99:

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan; 60%(0.60) of patients who are extracorporeal membrane oxygenation (ECMO) survivors are readmitted (Jen and Shew, Pediatrics 2010, 125: 1217-23).

100:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

101:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

102:

Patients with sepsis-induced tissue hypoperfusion can present with elevated blood lactate levels greater than 2 mmol/L(18.0 mg/dL) (Shankar-Hari et al., JAMA 2016, 315: 775-87).

103:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

104:

The initial vasopressor of choice for patients with septic shock is norepinephrine. Therapy should target a mean arterial pressure (MAP) of greater than or equal to 65 mmHg. Patients on vasoactive medication frequently have an

arterial line placed for blood pressure measurement (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Dellinger et al., Crit Care Med 2013, 41: 580-637).

105:

The 2021 Surviving Sepsis Campaign Guidelines suggest adjunct corticosteroids (typically intravenous hydrocortisone 200 mg daily in divided doses) for patients with septic shock who remain on vasopressors at a dose of 0.25 mcg/kg/min or higher of norepinephrine or epinephrine at least 4 hours after initiation (Evans et al., Crit Care Med 2021, 49: e1063-e143).

106:

Hospitalized patients with actual or suspected pelvic inflammatory disease (PID) should receive IV anti-infective therapy promptly. Patients can be transitioned to an oral anti-infective regimen 24 to 48 hours after demonstrating clinical improvement (Workowski et al., MMWR Recomm Rep 2021, 70: 1-187; Curry et al., Am Fam Physician 2019, 100: 357-64).

107:

Appendicitis during pregnancy is the most common non-obstetric surgical emergency. Diagnosis of appendicitis in a pregnant patient cannot be achieved with symptoms only. Further workup with laboratory testing is required. Pregnant patients will require surgery, as non-operative management is not recommended at this time (Di Saverio et al., World J Emerg Surg 2020, 15: 27).

108:

Hospital acquired refers to a condition that developed during the patient's current hospitalization. Criteria for hospital acquired conditions cannot be applied to patients who have been either discharged to home or transferred from a post-acute facility.

109:

Patients with chronic renal failure, immunocompromised state due to immunosuppressant therapy should be at the Acute level of care due to a higher risk of requiring emergency surgery on the index admission and high risk of recurrence after history of nonoperative management of a prior diverticulitis infection (Sartelli et al., World J Emerg Surg 2020, 15: 32).

110:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

111:

A patient recovering from appendicitis will be afebrile, have adequate pain management and be able to tolerate oral fluids (LoVecchio, N Engl J Med 2021, 384: 879).

112:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

113:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

114:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

115:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

116:

Fluid is replaced based on measured and insensible losses. Examples of measurable fluid loss include:

- Urine
- Nasogastric
- Wound
- Ostomy drainage
- Diarrhea
- Vomiting
- Fluid sequestration

Examples of insensible loss include:

- Perspiration

- Breathing.

The intent of the criteria is to address patients who have excessive fluid loss due to diarrhea and/or inability to take adequate oral fluids to replace fluid loss. If the losses are due to diarrhea, there is no expectation that the order for intravenous fluids will specify measurement of the diarrhea, replacement on a cubic centimeter (cc) for cc basis, or any titration of the fluid rate based on the number of cc's lost. Continuous intravenous fluid should be ordered in a clinically significant amount that indicates a need for replacement along with documentation of excessive diarrhea (to support that there are "losses").

117:

Resolution of infection is indicated by a decrease in temperature, a normalizing of white blood cells and other lab values, and improving imaging studies. If there is a wound present, signs of infection should be resolving (e.g., decrease in wound drainage, less pain at site).

118:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

119:

Instruction: To satisfy this criteria point, nonoperative management of appendicitis criteria was met on Episode Day 1 (patient diagnosed with appendiceal abscess or phlegmon) or the patient does not meet responder criteria and has been diagnosed with appendiceal abscess or phlegmon by imaging requiring nonoperative management since admission.

120:

Instruction: To satisfy this criteria point, the patient does not meet the responder criteria and pregnancy criteria had to be met on Episode Day 1.

121:

Instruction: To satisfy this criteria point, the patient does not meet responder criteria and severe illness criteria (temperature > 101°F(38.3°C) or protracted vomiting) had to be met on Episode Day 1 or have been diagnosed since admission.

122:

Diagnosis of tubo-ovarian abscess (TOA) can be made by several modes of imaging:

- Ultrasound (US)
- Computed tomography (CT)
- Scintigraphy

The preferred treatment of tubo-ovarian abscess (TOA) is intravenous broad-spectrum anti-infectives and in most cases, drainage. If medical management alone is not sufficient and symptoms persist or worsen (within 72 hours), then abscess drainage (ultrasound-guided or CT-guided drainage) is required (if surgical drainage is required, use General Surgical subset) (Goje et al., J Minim Invasive Gynecol 2021, 28: 556-64; Zhu et al., J Obstet Gynaecol 2021, 41: 1097-101).

123:

Occasionally, the abscess may evolve and require additional drain placement or require drain repositioning or upsizing (Thompson and Hrabe, Gastroenterol Clin North Am 2021, 50: 475-88).

124:

Drain manipulation includes:

- Placing an additional drain
- Repositioning existing drain
- Placing a larger drain

125:

While an elevated ammonia level is the classic laboratory finding used to diagnose hepatic encephalopathy, serial ammonia measurements are inferior to clinical assessment in gauging improvement or deterioration in a patient receiving treatment (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47).

126:

Instruction: Patients with end-stage liver disease are generally on chronic outpatient lactulose, rifaximin, or neomycin at home to control symptoms of hepatic encephalopathy (HE). Admission to a skilled nursing facility (SNF) may be appropriate to initiate or adjust the dosage of these medications due to worsening hepatic encephalopathy.

127:

Oral branched-chain amino acid (BCAA) enriched formulations may improve the manifestations of occult or minimal episodic hepatic encephalopathy (HE). Intravenous administration is ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47; J Hepatol 2014, 61: 642-59). Although BCAA has a beneficial effect on HE, there is no evidence regarding an effect on mortality, quality of life, or nutritional parameters (Gluud et al., Cochrane Database Syst Rev 2017, 5: CD001939).

128:

Lactulose is recommended by the American Association for the Study of Liver Diseases/European Association for the Study of the Liver guidelines as the first-line treatment for episodes of hepatic encephalopathy. It is a nonabsorbable substance used to treat an elevation in blood ammonia (NH₃) and reduces plasma ammonia concentration by converting soluble ammonia into insoluble ammonia in the colon. The laxative effect of lactulose results in the expulsion of trapped ammonium in the form of diarrhea. Lactulose can be administered orally, via a nasogastric tube, or as a retention enema (Swaminathan et al., Hepat Med 2018, 10: 1-11; J Hepatol 2014, 61: 642-59).

129:

Evidence has shown that L-ornithine L-aspartate (LOLA) is effective in lowering ammonia and improving mental status in patients with cirrhosis and hepatic encephalopathy (HE), with the oral form of LOLA being especially beneficial (Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13). In a 2017 randomized, controlled trial, patients with persistent HE showed improvement from intravenous LOLA administration, whereas oral supplementation was determined to be ineffective (Ferenci, Gastroenterol Rep (Oxf) 2017, 5: 138-47). A recent systematic review reassessed both oral and intravenous (IV) LOLA for the treatment of low-grade or minimal hepatic encephalopathy (MHE) and high grade or overt hepatic encephalopathy (OHE). The relative efficacy of LOLA depended on the type of HE as well as the route of administration. Patients with MHE, defined as patients with cirrhosis and alterations in mental status diagnosed by psychometric testing, appear to benefit more from the oral administration of LOLA than patients with high grade overt HE (Weissenborn, Drugs 2019, 79: 5-9; Butterworth et al., J Clin Exp Hepatol 2018, 8: 301-13).

130:

The renal and ototoxicity of neomycin have rendered this medication nearly obsolete in the treatment of hepatic encephalopathy (Tabbers et al., J Pediatr Gastroenterol Nutr 2014, 58: 258-74).

131:

Rifaximin is an anti-infective recommended by evidence-based clinical guidelines as an effective add-on therapy to lactulose for the treatment of overt hepatic encephalopathy (OHE). It has been shown to reduce healthcare resource utilization over the long term by lessening OHE recurrences and rehospitalization (Hudson and Schuchmann, European journal of gastroenterology & hepatology 2019, 31: 434). In a systematic review comparing the effectiveness and safety of interventions in patients with OHE, Rifaximin showed the greatest reduction in blood ammonia concentration (Zhu et al., Aliment Pharmacol Ther 2015, 41: 624-35).

132:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

133:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

134:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

135:

An ileus is a blockage, partial or complete, or paralysis of the bowel that prevents the passage of intestinal contents from moving forward. It is more common in the small bowel, as opposed to large bowel. It is often the result of surgical adhesions or hernia. Small bowel ileus symptoms typically present as acute and severe, with complaints of nausea and vomiting, cramps, bloating, and the retention of stool and flatus. It can often be treated medically. Large bowel ileus usually has a milder presentation and often is due to cancer. It may also be the result of adhesions or stenoses from recurrent diverticulitis or, less commonly, is associated with a sigmoid volvulus or hernia (Vilz et al., Dtsch Arztebl Int 2017, 114: 508-18).

136:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

137:

Hepatorenal syndrome (HRS) has been defined as acute renal failure in the presence of cirrhosis that manifests in profound circulatory dysfunction. A recent update in the diagnosis of HRS focuses on changes in serum creatinine levels instead of a fixed high serum creatinine value. Standard treatment for HRS includes drugs that increase blood pressure along with the albumin, a protein that is a volume expander. Per HRS guidelines, terlipressin with albumin is the recommended treatment, although in the U.S., because the use of terlipressin has not yet been approved, octreotide and midodrine are used in combination with albumin. A Cochrane Database Systematic Review analyzing the benefit or harmful effects of terlipressin given in combination with albumin found insufficient evidence in several small randomized controlled trials to support or refute terlipressin's superiority. Terlipressin was also associated with an increased risk of abdominal pain or diarrhea (Israelsen et al., Cochrane Database Syst Rev 2017, 9: Cd011532). Norepinephrine (also named noradrenaline) with albumin has been studied in relation to terlipressin and albumin and found to have a comparable efficacy. The use of norepinephrine in adult patients was validated by InterQual® consultants. Both regimens showed about a 40-50% response rate and up to a 50% rate of relapsing post-treatment. There is a high morbidity and mortality associated with HRS. Liver transplant screening is recommended for patients with this syndrome (Piano et al., Hepatol Int 2018, 12: 122-34; Acevedo and Cramp, World journal of hepatology 2017, 9: 293).

138:

Findings of hemodynamic instability include a sustained heart rate > 120/min, systolic blood pressure < 90 mmHg, systolic blood pressure < 110 mmHg with chronic hypertension, decrease in systolic blood pressure ≥ 30 mmHg from baseline, or shock by pulmonary embolism.

139:

While continuous renal replacement therapy represents over a third of dialytic interventions prescribed, to date there is insufficient evidence to determine its superiority over intermittent dialysis. Continuous dialysis may be

beneficial in hemodynamically unstable patients who are unable to tolerate the rapid fluid shifts associated with intermittent dialysis (Dellinger et al., Crit Care Med 2008; 36(1): 296-327; Crowley and Peixoto, Clin Chest Med 2009, 30: 29-43, vii-viii).

140:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

141:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

142:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

143:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

144:

Instruction: This criteria point can be selected only for patients who do not meet the responder criteria and were diagnosed on Episode Day 1 with diverticulitis and has a high risk comorbid condition (end stage liver disease, immunocompromised, or renal disease and chronic dialysis) or has since been diagnosed with a high risk comorbid condition and diverticulitis since admission.

145:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection.

To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: $1,500/\text{cu. mm} (1.5 \times 10^9/L)$

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu. mm} (1.5 \times 10^9/L)$

$= 330/\text{cu. mm} (330 \times 10^6/L)$

Normal range: $1,500-8,000/\text{cu. mm} (1.5-8.0 \times 10^9/L)$

An ANC of less than $500/\text{cu. mm} (500 \times 10^6/L)$ is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

146:

Abscesses include:

- Intra-abdominal
- Pelvic

147:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

148:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

149:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

150:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the

patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

151:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

152:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

153:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

154:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

155:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

156:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

157:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

158:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

159:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

160:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

161:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

162:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

163:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

164:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal

insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

165:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

166:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

167:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).

Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.